DESIGN THINKING FOR INNOVATION & ENTREPRENEURSHIP (DTIE)

Smart Classroom and Timetable Scheduler

One intelligent platform to schedule, allocate and keep every classroom, faculty member, student and parent in sync.



Under the supervision of Dr. G. Sudhakar
Associate Professor · Dept. of CSE · Subject: DTIE – Projects

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Vision and Mission

VISION

A connected campus where timetables are conflict-free, classrooms are used well, and every student, teacher and parent always knows what is happening next.

MISSION

Automate
Generate timetables automatically from faculty, subject, section and room data.

Prevent clashes
Detect and block faculty and classroom conflicts before they reach anyone.

Inform every role
Give students, faculty, parents and admins their own dashboard.

Stay flexible
Make timetable changes quick and easy to manage.

Problem Statement

Classrooms, faculty, subjects, students and timetables are managed in disconnected ways — so timetables are confusing or keep changing, rooms and faculty clash, and attendance, assignment and exam updates are missed.

15
pain points across 3 user groups

Students

- Difficulty finding classrooms
- Confusing or changing timetables
- Tracking assignments and deadlines
- Keeping track of attendance and exams
- Missing important notifications

Faculty

- Managing substitute classes
- Problems when unexpected leave occurs
- Managing timetables and room allocation
- Tracking student attendance and progress
- Managing assignments and projects

Parents

- Limited visibility of their child’s schedule
- Difficulty tracking attendance and academics
- Managing fee payments
- Keeping track of timetables and extra classes
- Missing important notifications

Research / Facts / Market & Industry Study

3
user groups studied

15
pain points identified

5
core requirements

4
role-based dashboards

CORE REQUIREMENT

Develop a centralized platform that manages classrooms, faculty, subjects, students and timetables.

- Automatically generate timetables
- Prevent faculty and classroom conflicts
- Improve classroom utilization
- Provide role-based dashboards
- Make timetable changes easy to manage

Real-time feedback & assessment

Technology provides instant feedback, helping both teachers and students understand progress and areas of improvement quickly.

Cost efficiency

Digital platforms can help reduce overall education costs by minimizing the need for printed books and transportation.

Target User – Customer

Student dashboard

- Timetable
- Attendance
- Assignments
- Exams
- Notifications

Faculty dashboard

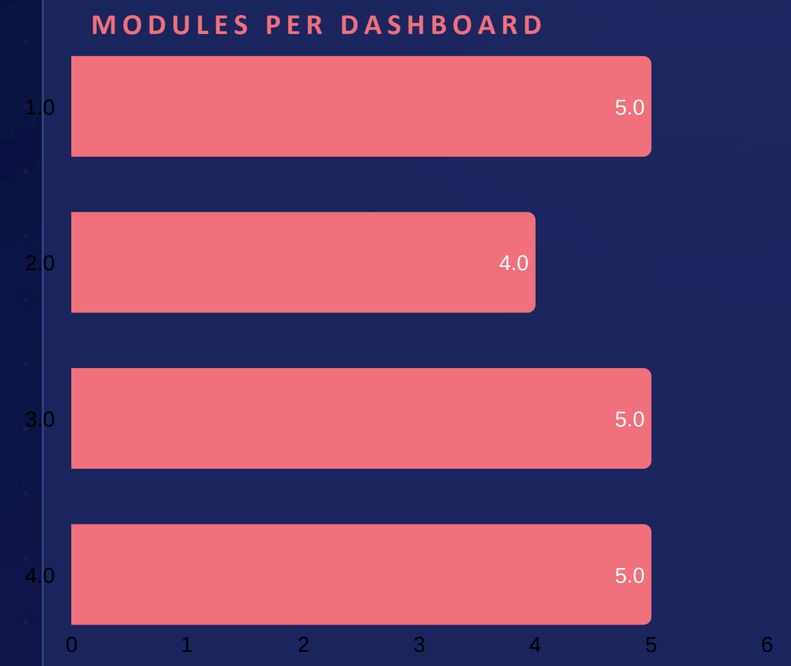
- Class schedule
- Announcements
- Attendance management
- Assignments

Parent dashboard

- Child’s timetable
- Attendance
- Academic performance
- Exam schedule
- Important alerts

Admin dashboard

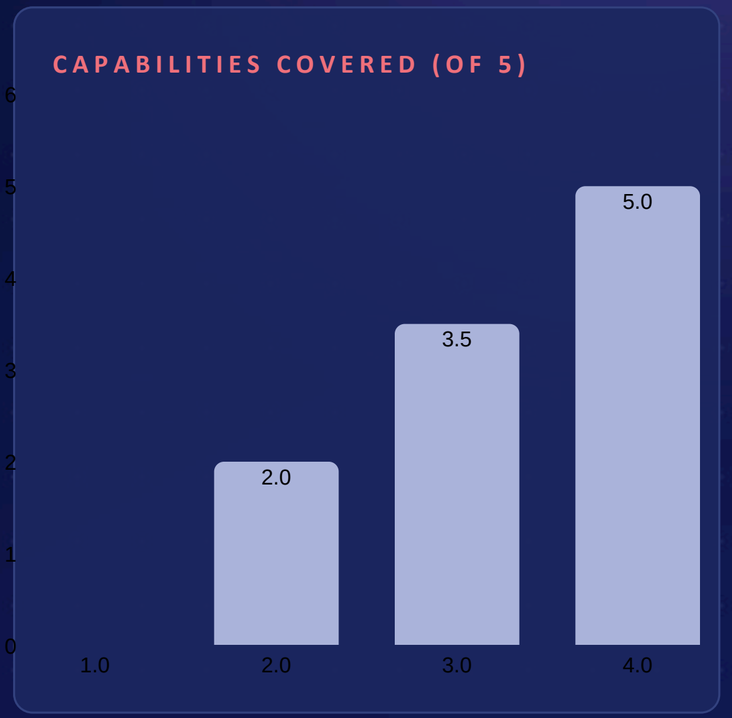
- User management
- Faculty management
- Subject management
- Classroom management
- Timetable management



19 dashboard modules serve four distinct roles from one shared data source.

Competitors / Benchmarks

Capability	Manual / sheets	Timetable tools	Campus ERP / LMS	Our platform
Automatic timetable generation	No	**Yes**	Partial	**Yes**
Conflict detection	No	**Yes**	Partial	**Yes**
Role-based dashboards	No	No	**Yes**	**Yes**
Attendance & notifications	No	No	**Yes**	**Yes**
Parent visibility	No	No	Partial	**Yes**



Our edge: one integrated platform that combines scheduling, classroom management and four role-based dashboards. Ratings are indicative at category level (Yes = 1, Partial = 0.5).

Risks and Gap

THE GAP

No single platform joins timetable generation, classroom availability and role-based visibility — so clashes, last-minute changes and missed updates reach users first.

RISK 01

Incomplete or wrong input data

MITIGATION

Validate faculty, room and period data at entry.

RISK 02

Unexpected faculty leave

MITIGATION

Support quick substitution and easy timetable modification.

RISK 03

Low adoption across roles

MITIGATION

Keep dashboards simple and specific to each user.

RISK 04

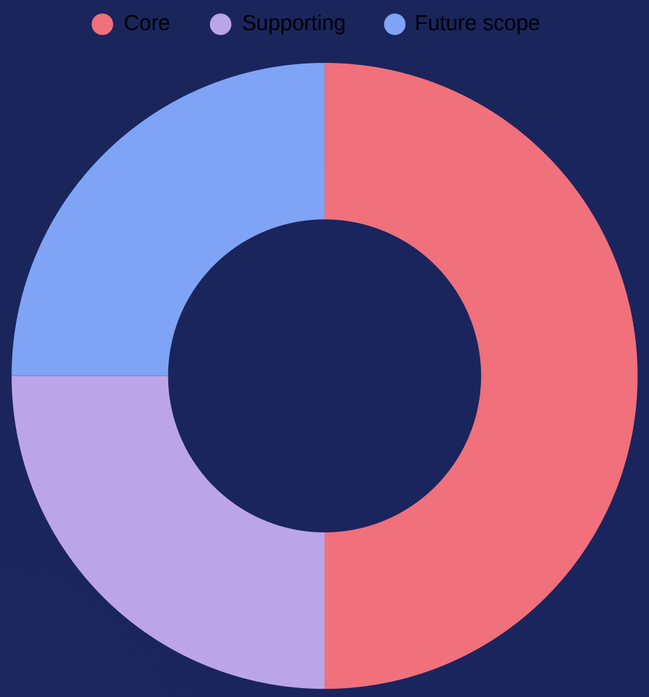
Scope creep

MITIGATION

Multi-voting keeps Core first; AI and mobile stay future scope.

Prioritising Ideas: Multi Voting

FEATURES BY PRIORITY



8 ideas shortlisted

Feature	Priority
Automatic timetable generation	Core (6/6)
Conflict detection	Core (6/6)
Role-based dashboards	Core (6/6)
Classroom management	Core (6/6)
Attendance management	Supporting (5/6)
Notifications	Supporting (5/6)
AI-based optimization	Future scope (4/5)
Mobile application	Future scope (4/5)

Concept / Idea / Prototype / Product / Solution

An efficient timetable generated while satisfying classroom, faculty and section constraints.

MODULE 1 · TIMETABLE

- Automatic generation
- Daily & weekly schedules
- Timetable modification

MODULE 2 · SMART CLASSROOM

- Classroom availability
- Room capacity
- Lab requirements
- Equipment information

Weekly Timetable · CSE Section A — 0 conflicts

	MON	TUE	WED	THU	FRI
P1	DBMS	OS	CN	SE	AI
P2	OS	CN	DBMS	LAB	SE
P3	CN	AI	LAB	DBMS	OS
P4	SE	DBMS	OS	AI	CN
P5	LAB	LAB	AI	OS	DBMS

Sample view: every slot respects faculty, room and section constraints.

Images / Storyboard / Architecture

Input

- Faculty details
- Subjects
- Student sections
- Classrooms and labs
- Room capacity
- Period timings
- Faculty availability

Processing

- Checks availability
- Assigns classrooms
- Allocates faculty
- Detects conflicts

Output

- Conflict-free timetable
- Classroom allocation
- Student schedule
- Faculty schedule
- Parent information

ROLE-BASED DASHBOARDS

Admin | Faculty | Student | Parent

Lean Canvas Model

PROBLEM

Confusing, changing timetables
Faculty and classroom clashes
Missed attendance and notifications

SOLUTION

Auto timetables with conflict detection

KEY METRICS

Conflicts prevented
Classroom utilization

UNIQUE VALUE

One platform for classrooms, faculty, subjects, students and timetables — conflict-free and easy to change.

UNFAIR ADVANTAGE

Scheduling plus four role dashboards

CHANNELS

Adoption through institution administration

CUSTOMER SEGMENTS

Students
Faculty
Parents
Administrators

COST STRUCTURE

Development, hosting and maintenance — offset by less printed material and transport.

REVENUE STREAMS (PROPOSED)

Institutional licence or subscription for colleges and schools.

Capital (Financial)

Digital platforms can reduce overall education costs by minimizing the need for printed books and transportation.

WHERE CAPITAL GOES

- Platform development and testing
- Hosting and infrastructure
- Maintenance and support
- Training for admins and faculty

WHERE VALUE RETURNS

- Less printed material and fewer books
- Reduced transportation needs
- Less manual scheduling effort
- Better classroom utilization

Roadmap / Timeline



SMART CLASSROOM AND TIMETABLE SCHEDULER

Thank you

- Conflict-free timetables
- Smart classroom allocation
- A dashboard for every role

